Multiple regression

Consider the following subset of a larger dataset:

wind_direction	home_age	pb_ppm
upwind	38	155
upwind	55	245
upwind	82	310
downwind	100	395
downwind	75	290
downwind	85	500

In the context of the Exide plant case study, which variable represents each of the following?

Continuous response

Categorical predictor

Continuous predictor

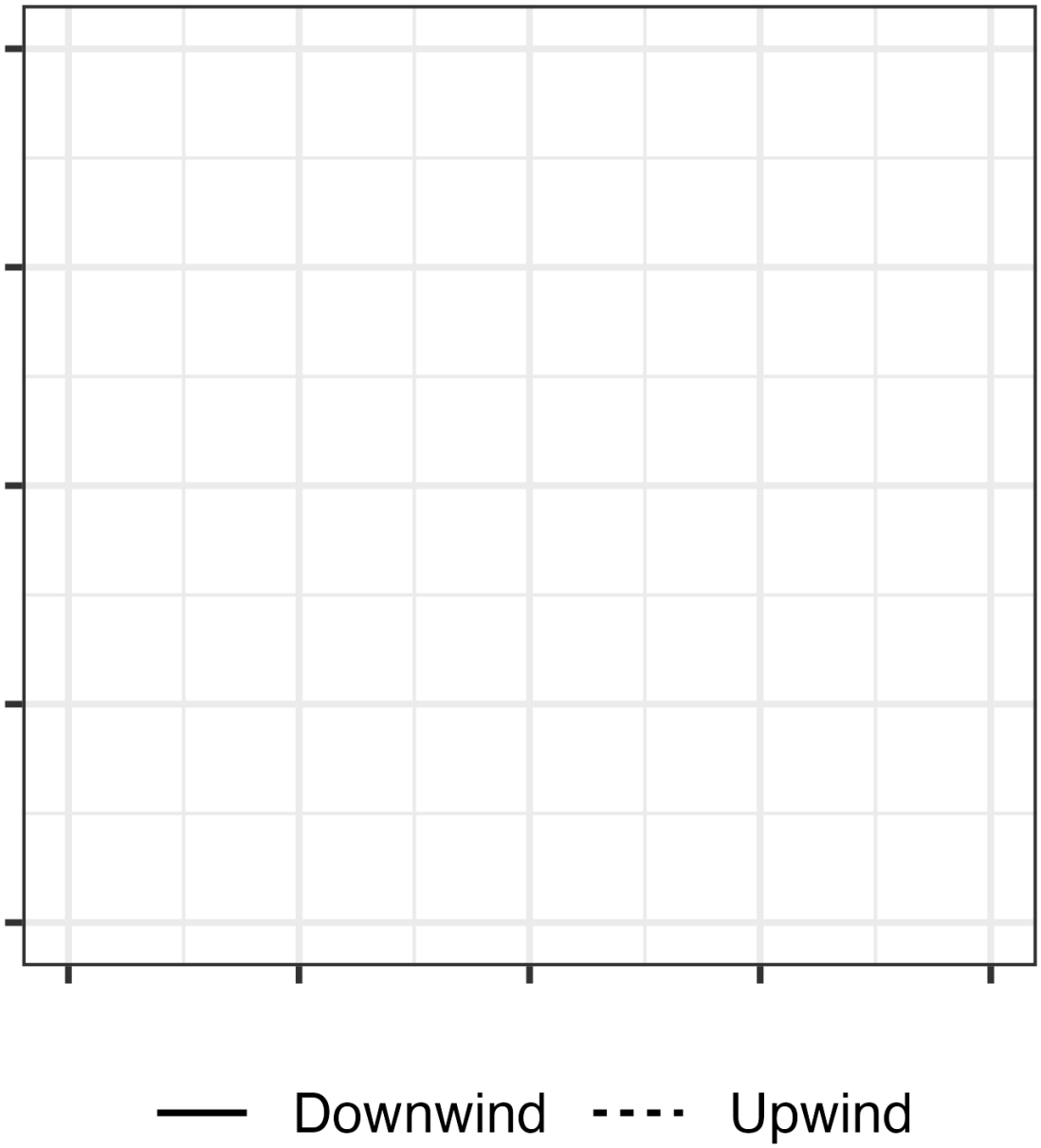
Fill in the blanks below to describe the multiple regression model in statistical notation.

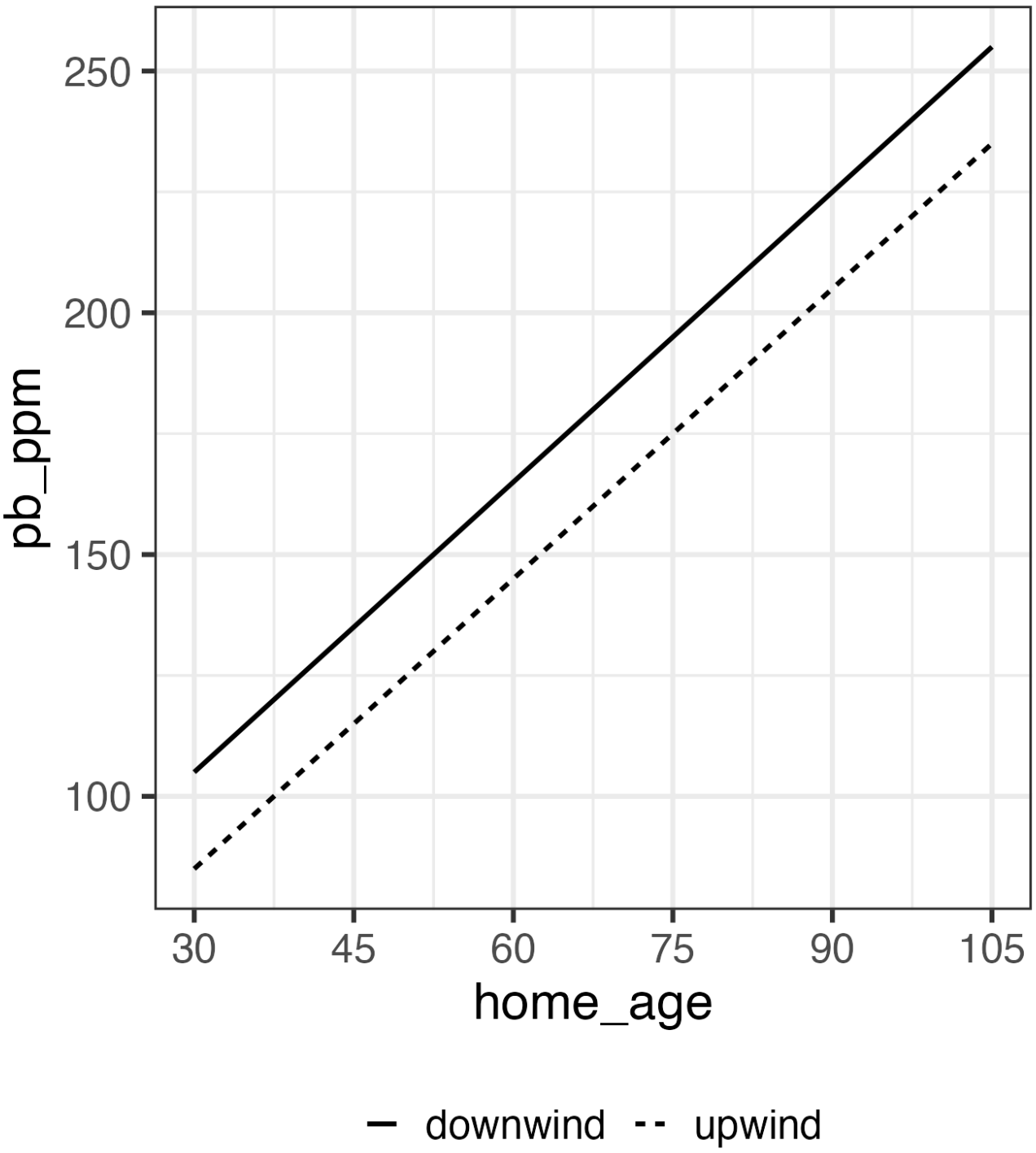
$$__ = __ + __ + __$$

$$__ \sim __(__, __)$$

Here, you'll draw lines representing the regression model.

1. Label the x- and y-axes with appropriate titles and tick labels.
2. Assume the following coefficient estimates: 45 for the intercept, -20 for "wind direction is upwind", and 2 for "home age".
3. Draw two lines: one for the downwind homes (solid) and one for the upwind homes (dashed).




250
200
150
100
pb_ppm
30
45
60
75
90
105
home_age
downwind
upwind